



Customer Shopping Behavior Analysis

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

DATASET OVERVIEW

Dataset Summary

The dataset contains 3,900 rows and 18 columns capturing customer demographics, purchase details, and shopping behavior. Missing data was limited to 37 values in the Review Rating column.

Customer Demographics

Age, Gender, Location, Subscription Status

Purchase Details

Item, Category, Amount, Season, Size, Color

Shopping Behavior

Discount, Promo Code, Previous Purchases, Frequency, Rating, Shipping

Exploratory Data Analysis in Python

We began with data preparation and cleaning in Python. The dataset was imported using pandas, with `df.info()` and `.describe()` used for initial exploration and summary statistics.

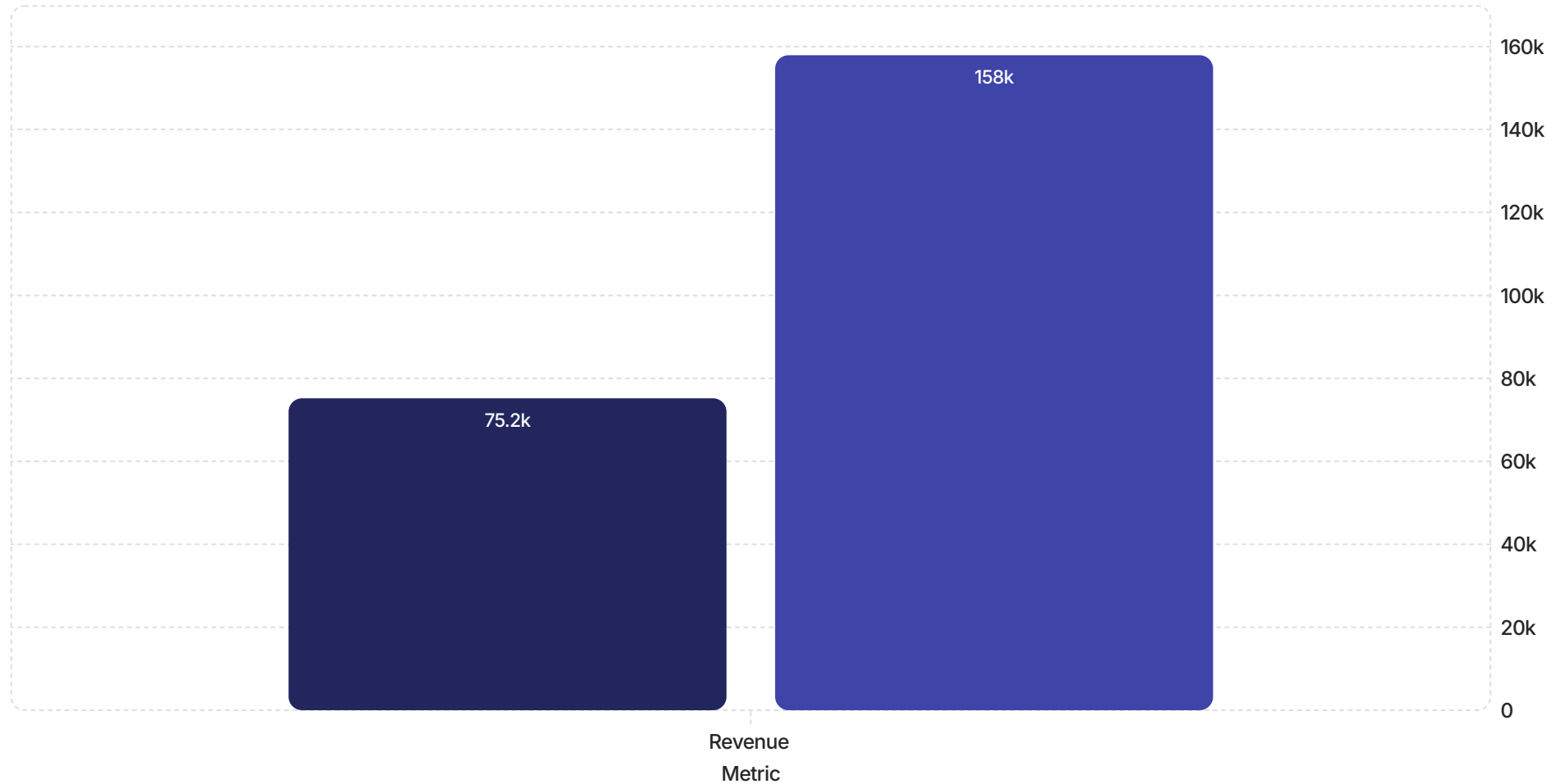
Missing values in the Review Rating column were imputed using the median rating of each product category. Columns were renamed to snake case for readability. Feature engineering created `age_group` and `purchase_frequency_days` columns. The redundant `promo_code_used` column was dropped after confirming it duplicated `discount_applied`. The cleaned DataFrame was then loaded into MySQL for SQL analysis.



Revenue & Spending Insights

Structured analysis in MySQL revealed key revenue and spending patterns across customer segments.

■ Female (\$) ■ Male (\$)



High-Spending Discount Users

839 customers used discounts yet still spent above the average purchase amount of \$59.76.

Shipping Comparison

Express shipping averaged \$60.48 per purchase vs. \$58.46 for Standard shipping.

Top Products by Rating & Discount Dependency

Products with the highest average review ratings and those most frequently purchased with discounts reveal where quality and promotions intersect.

Top 5 by Rating

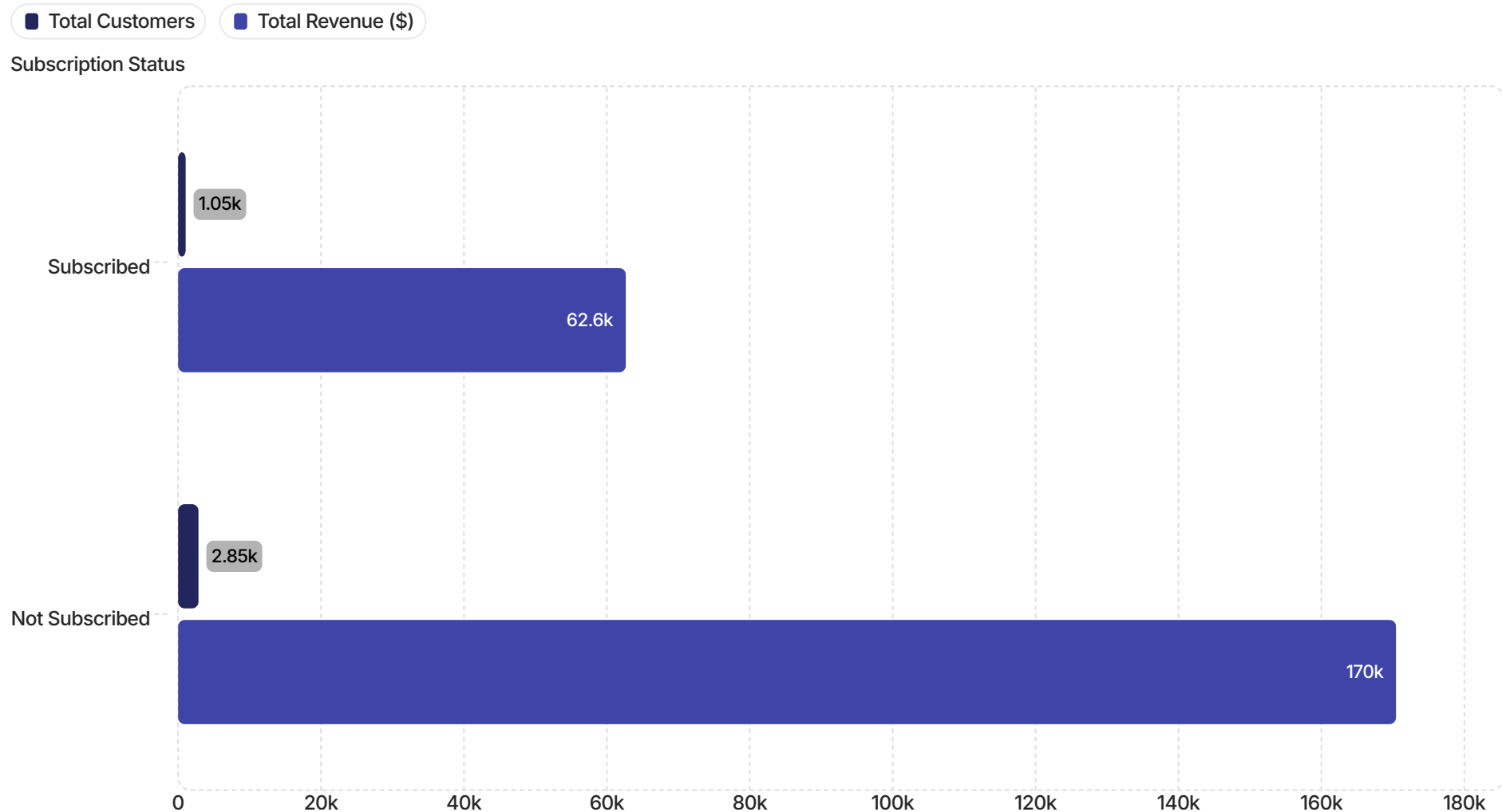
Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Subscribers vs. Non-Subscribers

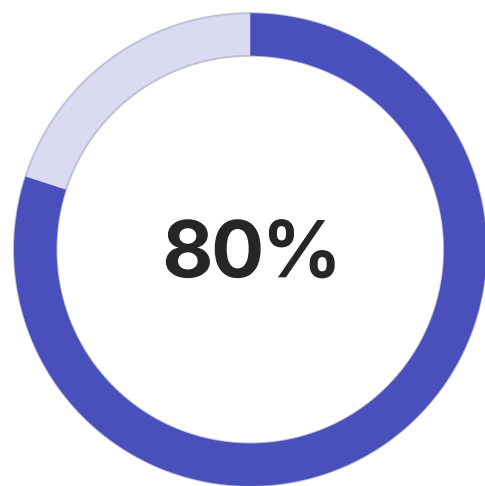
Subscription status reveals a significant revenue gap. Non-subscribers drive the majority of total revenue, while subscribers represent a smaller but engaged segment.



Average spend is nearly identical between groups - \$59.49 for subscribers vs. \$59.87 for non-subscribers suggesting subscription status does not significantly influence individual purchase size.

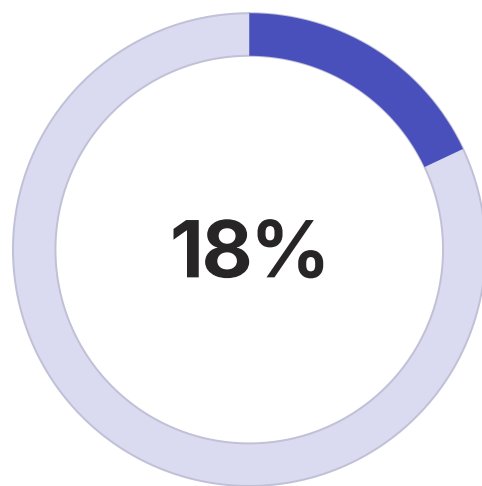
Customer Segmentation

Customers were classified into New, Returning, and Loyal segments based on purchase history. The vast majority fall into the Loyal category, indicating strong retention.



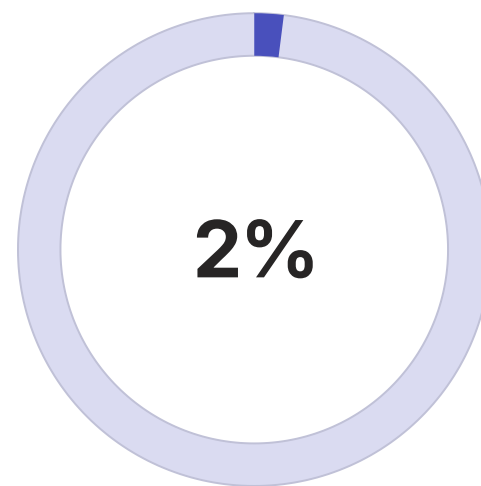
Loyal

3,116 customers with extensive purchase history



Returning

701 customers with repeat purchases



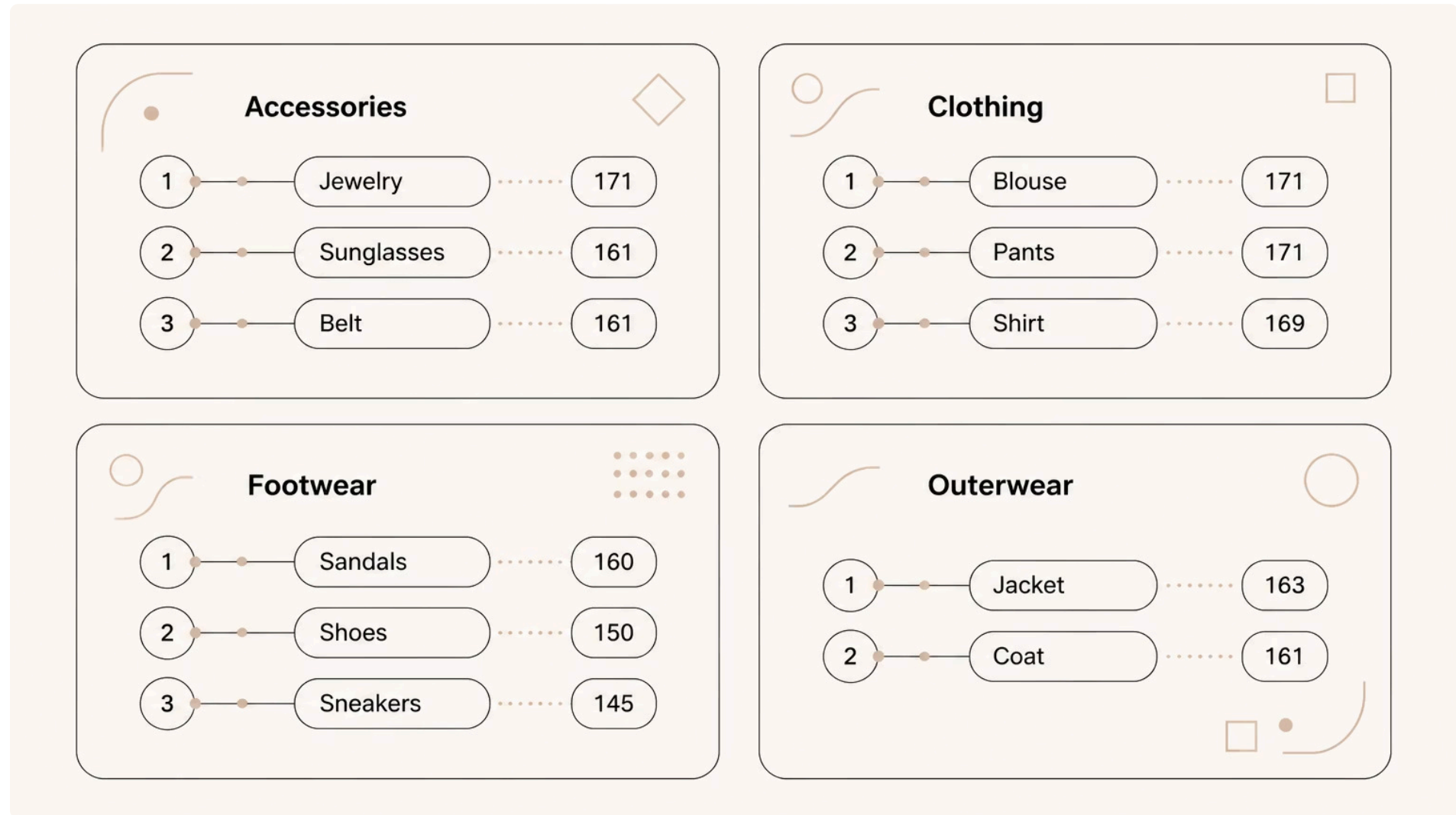
New

83 first-time customers

Among repeat buyers with more than 5 purchases, 2,518 remain non-subscribers while 958 have subscribed - suggesting repeat engagement does not strongly drive subscription conversion.

Top 3 Products per Category

The most purchased products within each category reveal clear customer preferences and potential focus areas for inventory and marketing.



Interactive Dashboard in Power BI

An interactive dashboard was built in Power BI to present insights visually. Key metrics include 3.9K total customers, a \$59.76 average purchase amount, and a 3.75 average review rating.

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

27%

Subscribed

The dashboard features filters for subscription status, gender, category, and shipping type, with visualizations for revenue by category and age group.



Business Recommendations

Based on the analysis, five strategic recommendations emerge to drive growth and customer engagement.



These recommendations target the largest opportunities: converting loyal non-subscribers, optimizing discount strategy, and focusing marketing on high-value segments.